

- Prompt 4: Now, check if these numbers are prime or not.

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Prompt 5: Great! The next set is a bit different. Are you familiar with the 24 game and its rules? If you are, I will give you 25 games to play (i.e., 25 sets of 4 digits to make 24 with).

(After an affirmative response from the agent and verifying that they know the rules)

Prompt 6: Perfect. You are welcome to provide the reasonings, but please remember to summarize all answers in a list following the order of the problems presented. Here is a set of 25 problems.

1. 1, 1, 1, 24
2. 1, 1, 2, 12
3. 1, 2, 3, 4
4. 1, 1, 4, 6
5. 1, 1, 3, 8
6. 0, 15, 39, 81
7. 3, 3, 4, 5
8. 2, 2, 2, 8
9. 6, 6, 8, 12
10. 1, 3, 3, 3
11. 4, 5, 9, 13
12. 1, 1, 1, 12
13. 5, 9, 12, 13
14. 6, 6, 8, 8
15. 4, 6, 7, 9
16. 7, 8, 9, 12
17. 2, 4, 6, 6
18. 6, 7, 8, 12
19. 1, 2, 3, 13
20. 3, 9, 12, 13
21. 1, 1, 6, 12
22. 1, 5, 6, 12
23. 1, 11, 12, 13
24. 2, 2, 8, 8
25. 2, 8, 8, 9

B Prompts Used in Study 2

Prompt 1: Are you familiar with the 24 game and its rules? If you are, I have 25 games for you to play (i.e., 25 sets of 4 numbers to make 24 with). First, state the rules briefly, and then start playing! One strict condition, though: You cannot write or use any code at all, even during your “thinking” processes. That would mean direct disqualification from the game.

1. 1, 1, 1, 24
2. 1, 1, 2, 12
3. 1, 2, 3, 4
4. 1, 1, 4, 6
5. 1, 1, 3, 8
6. 0, 15, 39, 81
7. 3, 3, 4, 5
8. 2, 2, 2, 8
9. 6, 6, 8, 12
10. 1, 3, 3, 3
11. 4, 5, 9, 13
12. 1, 1, 1, 12
13. 5, 9, 12, 13
14. 6, 6, 8, 8
15. 4, 6, 7, 9
16. 7, 8, 9, 12
17. 2, 4, 6, 6
18. 6, 7, 8, 12
19. 1, 2, 3, 13
20. 3, 9, 12, 13
21. 1, 1, 6, 12
22. 1, 5, 6, 12
23. 1, 11, 12, 13
24. 2, 2, 8, 8
25. 2, 8, 8, 9

Prompt 2: Perfect! Now, solve these 25 problems. Remember the condition: You cannot write or use any code at all, even during your “thinking” processes. That would mean direct disqualification from the game.

1. 7, 10, 12, 13
2. 4, 4, 8, 9
3. 2, 8, 10, 12
4. 2, 3, 9, 12
5. 5, 9, 12, 13
6. 3, 5, 9, 10
7. 2, 5, 7, 8
8. 1, 4, 9, 13
9. 5, 9, 10, 11
10. 2, 3, 8, 13